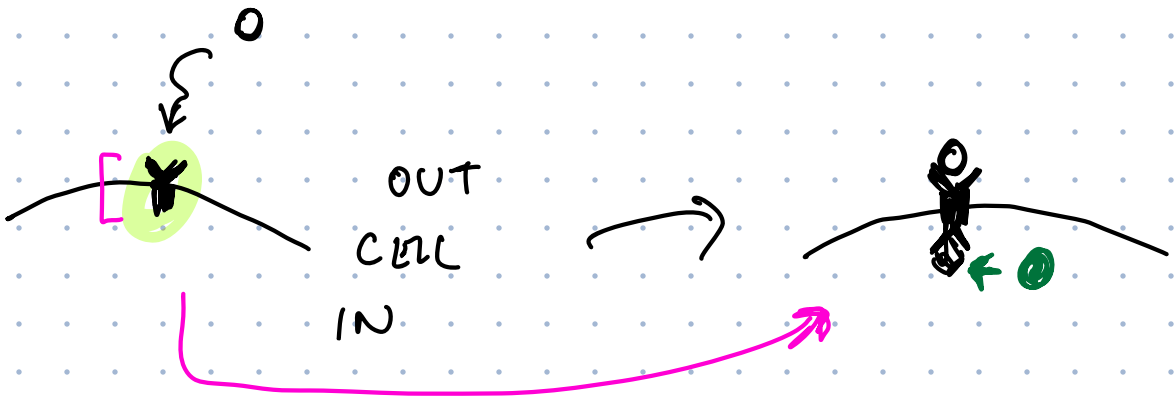
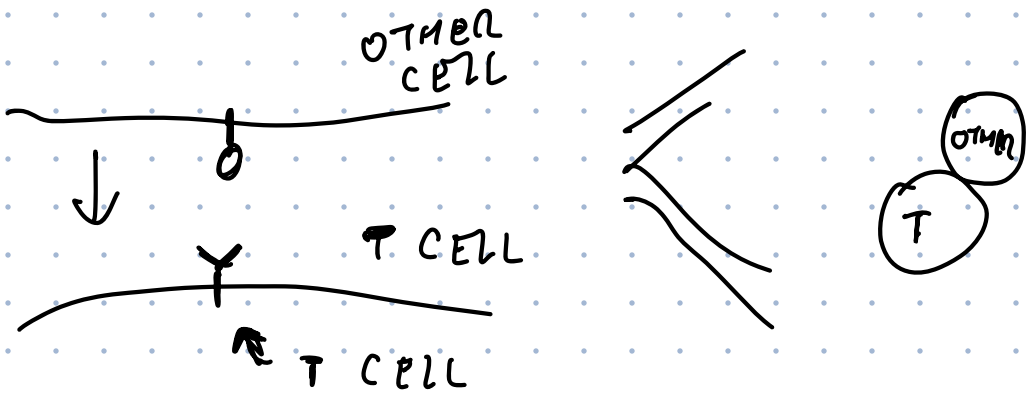
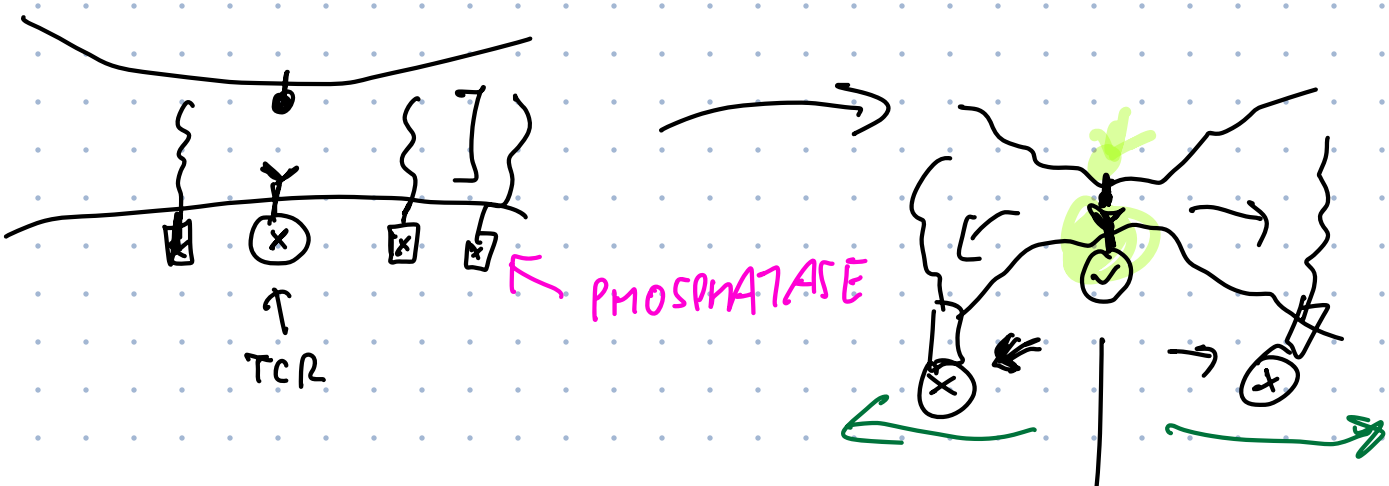


CONFORMATIONAL CHANGE

PS 5



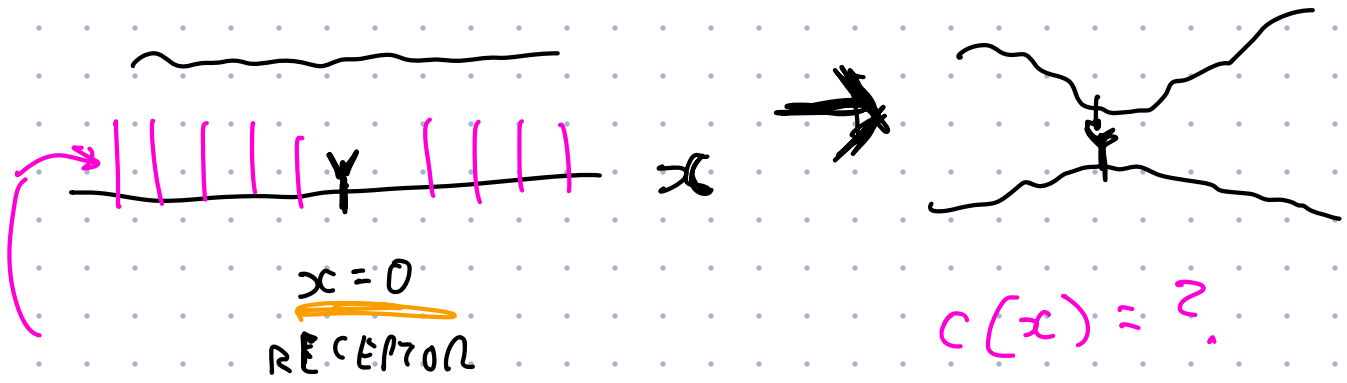
← NO CONFORMATIONAL CHANGE UPON BINDING



SEGREGATION

- HOW FAST DOES PHOSPHATASE SEGREGATE?
- HOW FAR?
- BY HOW MUCH? ☆ PS 5

BEFORE LIGATION



WHAT IS $c(0)/c_0$?

REVIEW:

DIFFUSION $J = -D\frac{\partial c}{\partial x}$ $\vec{J} = -D\nabla c$

$$\frac{\partial c}{\partial t} = -\frac{\partial J}{\partial x} \qquad \frac{\partial c}{\partial t} = -\nabla \cdot \vec{J}$$

ADVECTION $\quad J = +vc \qquad \vec{J} = +\vec{v}c$

BOTH DIFFUSION & ADVECTION

$$J = -D\frac{\partial c}{\partial x} + vc$$

$$\frac{\partial c}{\partial t} = -\frac{\partial}{\partial x}\left(-D\frac{\partial c}{\partial x} + vc\right)$$

$$= +\frac{\partial}{\partial x} D\frac{\partial c}{\partial x} - \frac{\partial (vc)}{\partial x}$$

IF $\quad v = -\mu \frac{\partial E}{\partial x}$

μ ↖ MOBILITY $\qquad \frac{\nu_M}{S} \quad \frac{N_M}{\rho N_{NM}}$

THEN (& D CONSTANT)

$$\frac{\partial c}{\partial t} = D\frac{\partial^2 c}{\partial x^2} + \mu \frac{\partial}{\partial x}\left(c\frac{\partial E}{\partial x}\right)$$

IF THERE IS NO FLUX AT BOUNDARIES,

AT STEADY STATE

$$c(x) = A\, e^{-\frac{\mu}{D}E(x)}$$

A: CONSTANT